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Roll No

EC-7003-CBGS

B.E. VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Optical Communication

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) List the various advantages and disadvantages of fibre optical communication over other communication technologies.
b) What do you understand by the modes in optical fibers? Compare single mode and multimode fiber.
2. a) Describe the advantage and disadvantages of optical fibre communication system over other terrestrial communication link.
b) A silica optical fiber with a core diameter large enough to be considered by ray theory analysis has a core refractive index of 1.50 and a cladding refractive index of 1.47.
Determine:
 - i) Critical angle at the core-cladding interface
 - ii) Numerical Aperture for the fiber
 - iii) Acceptance angle in air for the fiber

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3.
 - a) Discuss different types of fiber splicing with suitable diagram.
 - b) Write down short notes on optical fiber connectors.
4.
 - a) Discuss lens coupling of LEDs to optical fibers and outline the various techniques employed.
 - b) Explain the detection process in the $p-n$ photodiode. Compare this device with the $p-i-n$ photodiode.
5.
 - a) What is the significance of avalanche multiplication noise explain?
 - b) Write short notes on
 - i) Rise time budget
 - ii) Link power budget
6.
 - a) Discuss with a block diagram a coherent optical communication system.
 - b) Explain briefly the various key requirements needed in analyzing a link. Why do we require optical fiber link design.
7.
 - a) Explain the following requirements for the design of an optically amplified WDM link:
 - i) Link bandwidth
 - ii) Optical power requirements for a specific BER
 - b) What is the significance of Excited State Absorption (ESA) in erbium-doped fiber amplifier? Explain Excited State Absorption (ESA) with the help of suitable diagram.
8. Write notes on any two of the followings:
 - i) Optical Time-Domain Reflectometer
 - ii) WDM
 - iii) Coherent Detection
 - iv) Avalanche Photodiode

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